Documentation for BigQuery Fintech Dataset

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1 Preliminary Analysis of the Dataset

For this project, we make use of the BigQuery Fintech Dataset that is publicly available at https://www.kaggle.com/dataset fintech-dataset/data?select=loan.csv. We first analyse the `loans.csv` dataset.

1.1 `loans.csv` Dataset

The dataset consists of 270299 rows and 18 columns occupying space of 172MB. A following is the list of the columns together with their datatypes:

Column	Data Type	Description
`loan_id`	Identifier	A digit from 1 to 270299 that represents the unique identifier for the loan.
`customer_id`	Identifier	A unique string identifier that identifies a customer.
`loan_status`	Categorical	Consists of 7 unique categories: Current, Fully Paid, Charged Off, Late (31-120 days), In Grace Period, Late (16-30 days), Default.
`loan_amount`	Numerical	The principal amount loaned.
`state`	Categorical	The resident US state of the borrower, corresponding to all 51 US states.
`funded_amount`	Numerical	The principal amount loaned.
`term`	Categorical (Binary)	The term taken to pay off the loan. A binary column of either 36 or 60 months.
`int_rate`	Numerical	Interest rate charged on the loan.
`installment`	Numerical	The amount of installment paid per month in the term.
`grade`	Categorical	Investment grade of the loan ranging from A-G.
`issue_d`	Date	The issue date of the loan.
`issue_date`	Date	The issue date of the loan.
`issue_year`	Categorical	The issue year of the loan, ranging from all years between 2012 and 2019 inclusive.
`pymnt_plan`	Boolean	Unsure. This however was a boolean column
`type`	Categorical	The financing method of the loan. Consists of values like 'Individual', 'Joint' and 'DIRECT_PAY'
`purpose`	Categorical	The reason for obtaining the loan.
`description`	Categorical	The reason for obtaining the loan.
`notes`	Degenerate	Consisted of the values 'desc' only.

Table 1: Column analysis of `loans.csv`. Dtypes are those inferred by `pandas.read_csv` (pandas 3.0.3, in which text columns take the dedicated `str` dtype). [†]`issue_d` and `issue_date` are stored as text (`Jun-13` and `June 2013` respectively) and are read as `str`; the dtype shown is the result of parsing them with `pandas.to_datetime`.

Missing Values

The dataframe was free of missing values including placeholders for all columns except `description`.

Column	Column Data Type
loan_status	numerical
loan_amount	numerical
state	categorical
term	categorical (binary)
int_rate	numerical
installment	numerical
grade	categorical
issue_d	date
type	categorical
purpose	categorical

Duplicate Columns

Upon inspection of the columns, it is noted that the following pairs of columns are duplicated: `loan_amount, founded_amount` for the principal amount, `issue_date, issue_d` for the date of issue, `purpose, description` for the purpose of loan.

Constant Columns

Furthermore, the `pymnt_plan` nearly constant and consisting of 99.99% of `False` variables.

There were also type issues noted in the `type` column, entries like 'individual' and 'INDIVIDUAL' counted as unique entries in pandas and were subsequently merged into single entries.

To handle all the data quality issues identified, a new dataset called 'loan_cleaned.csv' was created by dropping the duplicate columns (`funded_amount`, `issue_d`, `description`), the non-varying columns (`notes`, `pymnt_plan`) and the identifier columns (`loan_id, customer_id`).[1], and fixing the entry inconsistencies seen in the `type` column.

The final selected columns together with their data category types are given:

[1]This step will be reversed in the future to allow for joining with the other dataset `customer.csv`.